



National Institute of Technology Rourkela
Department of Computer Science Engineering
Introduction to AI & ML (CS2011)
Mid-Semester Examination, B.Tech 2nd year

Duration: 2 hrs

Autumn 2025, 3rd Semester

Total marks: 30

Attempt all questions; ⟨CO-x: Course Outcome level⟩; ⟨BL-x: Bloom's taxonomy level⟩

1. (a) Define agent and environment. With suitable block diagram, define Simple Reflex Agents, Model based agent and Utility based agent. ⟨CO-1⟩ ⟨BL-1⟩ [5]
(b) A fraud detection dataset has 10,000 transactions, out of which only 100 are fraudulent. A classifier predicts all transactions as non-fraud. ⟨CO-2⟩ ⟨BL-2, 3⟩ [5]
 - i) Compute Accuracy, Precision, Recall, and F1-score.
 - i) Explain why Accuracy is misleading here.
2. (a) Consider the 8-puzzle problem with the following initial and goal states: ⟨CO-3⟩ ⟨BL-3⟩ [5]

Initial state:

2	8	3
1	6	4
7		5

Goal state

1	2	3
8		4
7	6	5

Figure 1: 8-puzzle problem scenario

Formulate the 8-puzzle problem as a state space representation using a 4-tuple. Draw the state space tree for the given problem, and determine the minimum number of moves required to reach the goal state using an optimal solution strategy.

- (b) Consider the following graph. Traverse the nodes of the graph using Breadth-First Search (BFS), taking A as the start state and H as the goal state. Show all intermediate steps of the traversal with the help of a suitable diagram, clearly illustrating

the queue implementation. Assume that whenever multiple nodes are available to be inserted into the queue, they are inserted in lexicographic order. <CO-3> <BL-3> [5]

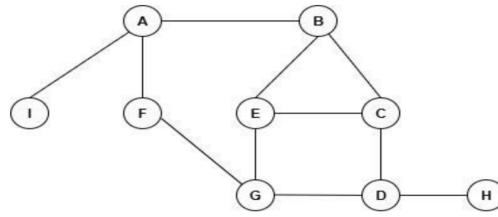


Figure 2: Graph traversal

3. (a) For a two-class classification problem with a one-dimensional feature set, the class conditional probability density functions of class and class are given below: <CO-3> <BL-3> [5]

$$p(x|w_1) = \begin{cases} (x-1)/4, & 1 \leq x \leq 3. \\ (5-x)/4, & 3 \leq x < 5. \end{cases} \quad (1)$$

$$p(x|w_2) = \begin{cases} 1/3, & 4 \leq x < 7. \end{cases} \quad (2)$$

The prior probabilities of class w_1 and class w_2 are $P(w_1) = 2/3$ and $P(w_2) = 1/3$ respectively. (i) Compute the total classification error for the Bayes classifier. (ii) Using the Bayes classifier, determine the class for $x = 2.5$.

- (b) (i) A binary classification model is being tested at three different decision thresholds: 0.9, 0.6, and 0.3. For each threshold, we record the confusion matrix values — True Positives (TP), False Positives (FP), False Negatives (FN), and True Negatives (TN) — as shown below: <CO-2, 3> <BL-3, 4> [3+2=5]

Threshold	TP	FP	FN	TN
0.9	32	8	18	142
0.6	44	20	6	130
0.3	50	35	0	115

Figure 3: Classification data

Plot the ROC curve using these points. Include the points (0,0) (always predict negative) and (1,1) (always predict positive) to complete the curve.

- ii) Write the PEAS description for an autonomous delivery drone operating in a smart city.